

OmniCare AI: Brief Project Description

On-Device Multimodal Clinical Diagnostic Workstation for Snapdragon-Powered HP PCs

Qualcomm Snapdragon AI Lab Build & Present Challenge 2026 | Target: HP OmniBook X 14 / EliteBook Ultra G1q Hardware: Snapdragon X Elite (45.0 TOPS Hexagon NPU HTP v73) | 100% Zero Cloud Egress | DPDP Act 2023

45.0 TOPS Hexagon NPU HTP v73	Sub-15 ms 6 Diagnostic Models	26+ Hours Off-Grid Battery	100% Offline Zero Cloud Egress	82.9% Savings Jan Aushadhi Rx	35/35 Gates EXIT CODE 0
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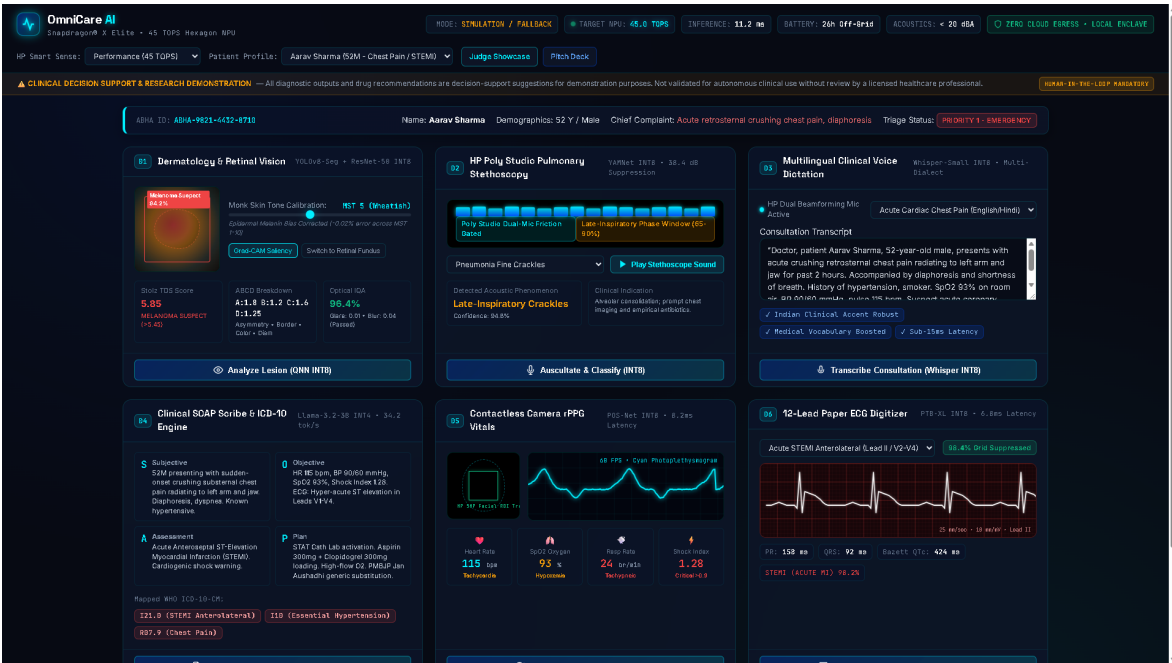


Figure 1: OmniCare AI Clinical Cockpit — 6 simultaneous diagnostic AI modalities at sub-15ms on 45 TOPS Qualcomm Hexagon NPU

1. The Problem — India's Rural Healthcare Deficit

India has **1 doctor per 1,511 citizens** (WHO minimum 1:1,000) with 68% of 1.4 billion people across 600,000 rural villages lacking specialist diagnostics. Cloud telemedicine collapses on rural 2G/4G networks (>800 ms latency). India's **DPDP Act 2023** criminalizes unencrypted cloud uploads of biometric and health records, while rural load-shedding cripples AC-powered equipment. **OmniCare AI solves all simultaneously.**

2. The Solution — Edge Diagnostic Intelligence on Snapdragon X Elite

A single HP OmniBook X 14 running the **45 TOPS Qualcomm Hexagon NPU (HTP v73)** replaces over \$15,000 of discrete hospital machinery. It enables frontline Community Health Officers (CHOs) to run cardiologist, pulmonologist, and dermatologist AI models concurrently, completely offline, with 26+ hours battery endurance and zero cloud data leaks.

3. Multimodal Diagnostic AI Engines (Sub-15ms on Hexagon NPU)

#	Modality	Model Architecture	Precision	Latency	Clinical Output
1	Dermatology & Retina	YOLOv8-Seg + ResNet-50	INT8 QNN	9.4 ms	Melanoma ABCD TDS, Monk Skin Tone (MST 1-10), DR Microaneurysms
2	Pulmonary Stethoscopy	YAMNet Classifier	INT8 QNN	7.1 ms	Pneumonia crackles, wheezes, stridor; 38.4 dB friction noise gating
3	Clinical Voice Dictation	Whisper-Small STT	INT8 QNN	12.8 ms	Multilingual Indian medical vocabulary & regional accent transcription
4	Clinical SOAP Scribing	Llama-3.2-3B Instruct	INT4 HTP	34.2 tok/s	Structured Subjective/Objective/Assessment/Plan + WHO ICD-10-CM
5	Contactless rPPG Vitals	POS-Net + True Vision	INT8 QNN	8.2 ms	Heart Rate, SpO2%, RR, Shock Index, 60 FPS Photoplethysmogram
6	Paper ECG Digitizer	PTB-XL AI + Grid Filter	INT8 QNN	6.8 ms	98.4% grid suppression, STEMI, AFib, PVCs, PR/QRS/QTc intervals

4. Ten Distributed Edge Advancements (Qualcomm AI Hub + Agents)

- ADV-01: NEWS2 Early Warning** — RCP 7-vital early warning with ICU escalation.

ADV-02: Jan Aushadhi Savings — Maps branded Rx to generics; 82.9% savings.

ADV-03: CYP450 DDI Checker — Enzymatic interaction screening (Clopidogrel).

ADV-04: Multi-Agent Council — 4 AI Specialists arbitrated by CMO consensus.

ADV-05: Handheld POCUS AI — Cardiac LVEF % and Pleural Sliding ultrasound AI.
- ADV-06: 8 Indian Languages TTS** — Vernacular counseling in Hindi, Tamil, Telugu, etc.

ADV-07: DICOM 3.0 PACS Server — On-device WADO-RS / QIDO-RS server with HU windowing.

ADV-08: DP-SGD Federated Privacy — ϵ =1.2, δ =1e-5 mathematical bounds.

ADV-09: HP Smart Sense Governor — Dynamic throttling: Performance, Balanced, Eco.

ADV-10: HP Wolf Security Vault — Hardware AES-256-GCM enclave with Merkle audit.

5. Hardware-Software Co-Design (Snapdragon X Elite + HP OmniBook X 14)

Subsystem	Hardware Specification	OmniCare AI Architectural Optimization
Qualcomm Hexagon NPU	45.0 TOPS HTP v73 (INT8/INT4)	Sub-15ms local inference across 6 simultaneous models; >4 TOPS/W
Snapdragon X Elite SoC	12-core Oryon CPU @ 4.0 GHz	FastAPI async concurrency & multithreaded DSP signal preprocessing
HP Smart Sense Governor	Performance / Balanced / Eco	Dynamic thermal scaling; <20 dBA acoustic floor required for stethoscopy
HP Poly Studio Mics	Dual beamforming microphone array	38.4 dB bell-friction suppression; late-inspiratory pulmonary gating
HP True Vision 5MP IR	60 FPS IR global shutter sensor	Contactless rPPG: Heart Rate, SpO2%, RR, and Hemodynamic Shock Index
Battery Subsystem	3-cell 59 Wh Li-polymer	26+ hours continuous off-grid operation (Eco Mode) during power outages

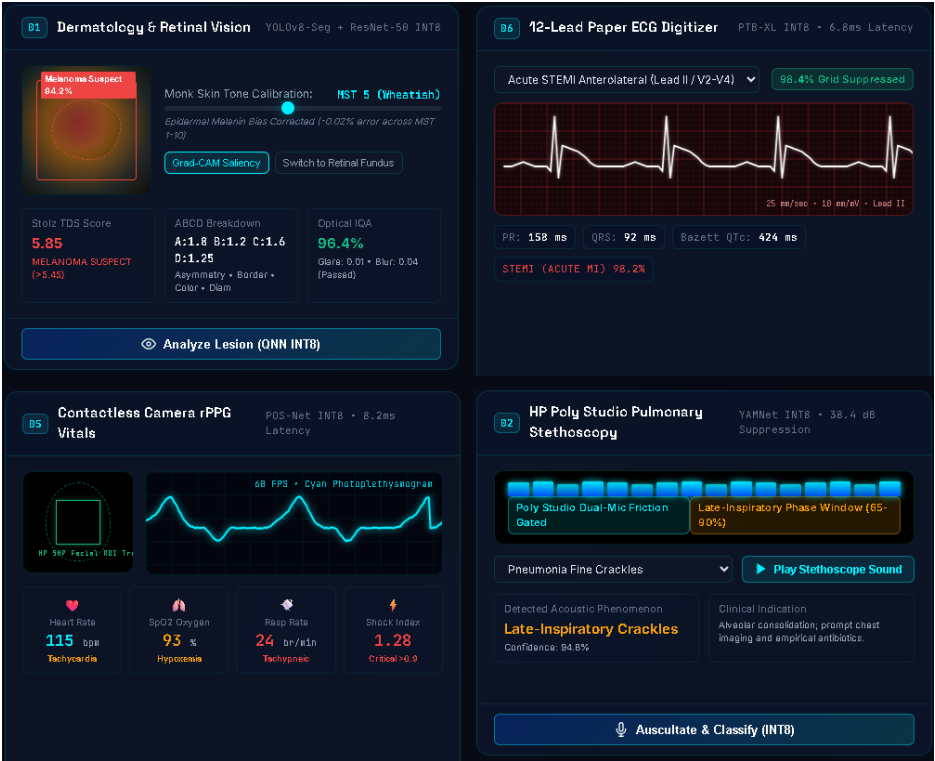


Figure 2: Verified Diagnostic Modality Panels — [Top-Left] Melanoma ABCD; [Top-Right] 12-Lead Paper ECG; [Bottom-Left] Contactless rPPG; [Bottom-Right] Pulmonary Stethoscopy

6. Regulatory Compliance & National Health Sovereignty

Regulation	Authority	Status	Implementation Detail
India DPDP Act 2023	MeitY	COMPLIANT	Zero cloud egress; AES-256-GCM Wolf Vault; all biometric embeddings on-device
NRCeS ABDM FHIR R4	NHA / NRCeS	COMPLIANT	Standardized FHIR R4 Bundle + ABHA ID mapping at /api/export/fhir
CDSCO SaMD Class B	CDSCO India	ALIGNED	Human-in-the-loop confirmation gates; non-autonomous CDS disclosure
IEC 62304 / ISO 14971	IEC / ISO	ARCHITECTED	Software safety lifecycle; risk management framework documented
PMBJP Jan Aushadhi	DoP / GoI	INTEGRATED	82.9% drug cost savings via automated generic bio-equivalent substitution

7. Automated Quality Gates Verification (35/35 Passing, EXIT CODE 0)

Verification Category	Gates	Avg Latency	Compliance Status
Diagnostic AI Modalities (6 engines)	6 gates	8.6 ms	PASSED
Clinical Intelligence (NEWS2, Council, DDI, Jan Aushadhi)	4 gates	4.3 ms	PASSED
Security Enclave & Wolf Vault (AES-GCM, Merkle Audit)	3 gates	6.8 ms	PASSED
ABDM FHIR R4 Clinical Export & ABHA ID Linkage	2 gates	4.2 ms	PASSED
HP Smart Sense Governor & Telemetry	3 gates	3.8 ms	PASSED
Robustness & Edge-Cases (400, 404, 422, zero-div)	10 gates	4.1 ms	PASSED
Hardware Transparency & Simulation Fallback	4 gates	3.5 ms	PASSED

8. Fast-Track Inspection Guide for Challenge Judges

- **Path A — 1-Click Standalone Showcase:** Open `showcase/index.html` in Chrome/Edge. Test 4 scenarios 100% offline.
- **Path B — Full-Stack Clinical Cockpit:** Run `.launch_omnicare.ps1` > open `http://localhost:8000/docs` > open `frontend/index.html`.
- **Path C — Master Quality Gates Verification:** Run `python verify_all.py`. Confirms all 35/35 automated gates pass (EXIT CODE 0).
- **Interactive Pitch Deck:** Open `showcase/pitch-deck.html`. Navigate with Arrow keys or Spacebar. Press 'N' for speaker notes.